

Data Structures

02

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Dense reference: mechanics, complexities, code, diagrams, and interview takeaways in one place.

1 Overview

Data structures are the substrate of every algorithm. Choosing the wrong one is the single most common reason candidates fail coding interviews — the algorithm is right, but $O(n)$ lookups in a list where a hash map would give $O(1)$ makes the solution too slow. This file is organized as a working reference: read top-to-bottom once, then use the complexity table and decision guide during revision.

Mental model for selection:

1. What operations do I need? (search, insert, delete, min/max, range, prefix?)
2. What are the frequency/order constraints? (random access? FIFO? LIFO? sorted?)
3. What are the acceptable complexities?
4. Then pick the structure.

2 Master Complexity Table

All complexities are average-case unless marked. W = worst case.

Data Structure	Access	Search	Insert	Delete	Space	Notes
Array (static)	$O(1)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	Insert/delete shifts elements
Dynamic Array (list)	$O(1)$	$O(n)$	$O(1)$ amort	$O(n)$	$O(n)$	Append amortized $O(1)$
String (immutable)	$O(1)$	$O(n)$	$O(n)$	$O(n)$	$O(n)$	Concat is $O(n)$; use StringBuilder
Hash Table	N/A	$O(1)$	$O(1)$	$O(1)$	$O(n)$	W: $O(n)$ with all collisions
Singly Linked List	$O(n)$	$O(n)$	$O(1)$ head	$O(n)$	$O(n)$	$O(1)$ if you have the node reference
Doubly Linked List	$O(n)$	$O(n)$	$O(1)$ ends	$O(1)^*$	$O(n)$	* $O(1)$ if node ref known
Stack	$O(n)$	$O(n)$	$O(1)$ push	$O(1)$ pop	$O(n)$	Top access $O(1)$
Queue	$O(n)$	$O(n)$	$O(1)$ enq	$O(1)$ deq	$O(n)$	Front access $O(1)$
Deque	$O(1)$ ends	$O(n)$	$O(1)$ ends	$O(1)$ ends	$O(n)$	Both ends $O(1)$
BST (balanced)	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(n)$	Unbalanced: $O(n)$ worst
BST (unbalanced)	$O(n)$ W	$O(n)$ W	$O(n)$ W	$O(n)$ W	$O(n)$	Degenerate = linked list
AVL Tree	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(n)$	Strict balance; fast lookup
Red-Black Tree	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(n)$	Looser balance; fast insert/delete
Binary Heap (min/max)	$O(1)$ top	$O(n)$	$O(\log n)$	$O(\log n)$	$O(n)$	Build heap: $O(n)$

Data Structure	Access	Search	Insert	Delete	Space	Notes
Trie	$O(m)$	$O(m)$	$O(m)$	$O(m)$	$O(n*m)$	m = key length
Graph (adj list)	$O(V+E)$	$O(V+E)$	$O(1)$	$O(E)$	$O(V+E)$	Sparse graphs
Graph (adj matrix)	$O(1)$	$O(1)$ edge	$O(1)$	$O(1)$	$O(V^2)$	Dense graphs
Union-Find	$O(\alpha(n))$	$O(\alpha(n))$	$O(\alpha(n))$	-	$O(n)$	α = inverse Ackermann, $\approx O(1)$
Segment Tree	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(n)$	Range queries + point updates
Fenwick Tree (BIT)	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(n)$	Simpler than segment tree
LRU Cache	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(n)$	Hash map + doubly linked list
Bloom Filter	N/A	$O(k)$	$O(k)$	N/A	$O(m)$	k =hash funcs, probabilistic

3 The Data Structures

3.1 Arrays & Dynamic Arrays

What it is: Contiguous block of memory where each element sits at a fixed offset from the base address. Random access is $O(1)$ because $\text{address} = \text{base} + \text{index} * \text{element_size}$.

Internal mechanics:

```
Static Array (size=6):
Index:  [0] [1] [2] [3] [4] [5]
Value:  [10] [20] [30] [40] [50] [60]
        ^               ^
        base addr       base + 5*sizeof(int)

Dynamic Array (Python list) - amortized resize:
size=4, capacity=4:  [1][2][3][4][ ][ ][ ][ ] ← capacity doubles on overflow
Append 5 → capacity=8:[1][2][3][4][5][ ][ ][ ]
Append cost analysis (n appends):
- n-1 cheap appends:  $O(1)$  each
-  $\sim \log(n)$  doublings: cost  $1+2+4+\dots+n = 2n$  copies total
- Amortized per append:  $2n / n = O(1)$  ✓
```

Complexity:

Op	Best	Avg	Worst
Access	$O(1)$	$O(1)$	$O(1)$
Search	$O(1)$	$O(n)$	$O(n)$
Append	$O(1)$	$O(1)$	$O(n)$
Insert	$O(1)$	$O(n)$	$O(n)$
Delete	$O(1)$	$O(n)$	$O(n)$

When to use:

- › Random access by index is needed
- › Data is mostly read; writes are appends
- › Cache locality matters (iteration speed)

When NOT to use:

- › Frequent mid-list insertions/deletions (use linked list or deque)

- › Need fast search (use hash map or BST)

Key Python patterns:

```

1 # Two-pointer on sorted array
2 left, right = 0, len(arr) - 1
3 while left < right:
4     s = arr[left] + arr[right]
5     if s == target: return [left, right]
6     elif s < target: left += 1
7     else: right -= 1
8
9 # Sliding window
10 window_sum = sum(arr[:k])
11 max_sum = window_sum
12 for i in range(k, len(arr)):
13     window_sum += arr[i] - arr[i - k]
14     max_sum = max(max_sum, window_sum)
15
16 # In-place reversal
17 arr[i], arr[j] = arr[j], arr[i]

```

Interview takeaway: Arrays are the default. When the problem says "return sorted" or "in-place", think array + two pointers. When you need to insert/delete mid-sequence often, question whether an array is right.

Real interview use-case: Two Sum (hash map), Sliding Window Maximum, Kadane's (max subarray), rotate array in-place, merge sorted arrays.

3.2 Strings

What it is: Sequence of characters. In Python and Java, strings are **immutable** — every concatenation creates a new object. In C++, `std::string` is mutable but each `+=` may reallocate.

Internal mechanics:

```

Python string "hello":
h | e | l | l | o
Each character is a Unicode code point (Python 3: variable width internally)
id("hello" + "x") != id("hello") ← new object every concat

String concatenation cost in a loop (naïve):
for i in range(n): result += s[i]
Iteration 1: copy 1 char → O(1)
Iteration 2: copy 2 chars → O(2)
...
Iteration n: copy n chars → O(n)
Total: O(1+2+...+n) = O(n²) ← DISASTER for large n

String Builder pattern:
parts = []
for c in chars: parts.append(c) ← O(1) amortized each
result = "".join(parts) ← O(n) one pass
Total: O(n) ✓

```

Complexity:

Op	Complexity	Note
Index access	$O(1)$	
Slice <code>s[i:j]</code>	$O(j-i)$	Creates new string
Concatenate	$O(n+m)$	Both strings copied

Op	Complexity	Note
Find/search	$O(n*m)$	Naive; KMP/Z gives $O(n+m)$
Comparison	$O(n)$	Character by character
Join	$O(n)$	One pass

Key Python patterns:

```

1 # Always join, never += in loop
2 result = "".join(list_of_chars)
3
4 # Reverse a string
5 rev = s[::-1]
6
7 # Check palindrome
8 def is_palindrome(s):
9     s = s.lower()
10    l, r = 0, len(s) - 1
11    while l < r:
12        while l < r and not s[l].isalnum(): l += 1
13        while l < r and not s[r].isalnum(): r -= 1
14        if s[l] != s[r]: return False
15        l += 1; r -= 1
16    return True
17
18 # Anagram check
19 from collections import Counter
20 Counter(s) == Counter(t)
21
22 # Sliding window on string (distinct chars)
23 from collections import defaultdict
24 freq = defaultdict(int)
25 l = 0
26 for r, c in enumerate(s):
27     freq[c] += 1
28     while len(freq) > k:
29         freq[s[l]] -= 1
30         if freq[s[l]] == 0: del freq[s[l]]
31         l += 1

```

Interview takeaway: Never concatenate strings in a loop. Use `"".join(parts)`. For pattern matching, think sliding window + frequency map. Anagram/permutation problems → character frequency counter.

Real interview use-case: Longest substring without repeating characters, group anagrams, valid palindrome, minimum window substring, encode/decode strings.

3.3 Hash Tables

What it is: Maps keys to values in $O(1)$ average by computing $\text{hash}(\text{key}) \% \text{capacity}$ to find a bucket, then storing the key-value pair there.

Internal mechanics:

Hash Table with Chaining (load factor = 0.75):

```

Buckets:
[0] → None
[1] → ("apple", 5) → ("grape", 2) → None ← collision! same bucket

```

```

[2] → ("banana", 3) → None
[3] → None
[4] → ("cherry", 7) → None
...

hash("apple") % 8 = 1
hash("grape") % 8 = 1 ← collision → chain extends
hash("banana") % 8 = 2

Open Addressing (linear probing):
[0]: empty
[1]: ("apple", 5)
[2]: ("grape", 2) ← collision at 1, probe to 2
[3]: ("banana", 3)
...

Load Factor ( $\alpha$ ) =  $n / \text{capacity}$ 
 $\alpha < 0.7$ : few collisions,  $O(1)$  expected
 $\alpha > 0.9$ : many collisions →  $O(n)$  worst
→ resize (rehash all) when  $\alpha$  threshold exceeded
Python dict resizes at 2/3 full → rehash to 2x capacity

Good Hash Function Properties:
1. Deterministic (same key → same hash)
2. Uniform distribution (minimize collisions)
3. Fast to compute
4. Avalanche effect (small key change → big hash change)

```

Collision resolution comparison:

Strategy	Pro	Con
Chaining	Simple, works at high load	Extra memory for pointers
Linear probing	Cache-friendly	Clustering degrades performance
Quadratic probe	Less clustering	May not probe all slots
Double hashing	Best distribution	Two hash functions needed

Complexity:

Op	Average	Worst
Search	$O(1)$	$O(n)$
Insert	$O(1)$	$O(n)$
Delete	$O(1)$	$O(n)$

When to use:

- › $O(1)$ lookup by key
- › Counting frequencies
- › Caching / memoization
- › Deduplication

When NOT to use:

- › Need sorted order (use BST/sorted array)
- › Keys not hashable (use tree map)
- › Memory is very tight

Key Python patterns:

```

1 from collections import defaultdict, Counter
2
3 # Frequency count
4 freq = Counter("anagram")          # Counter({'a': 3, 'n': 1, ...})
5
6 # Default dict avoids KeyError
7 graph = defaultdict(list)
8 graph['A'].append('B')
9
10 # Two Sum O(n)
11 def two_sum(nums, target):
12     seen = {}
13     for i, n in enumerate(nums):
14         complement = target - n
15         if complement in seen:
16             return [seen[complement], i]
17         seen[n] = i
18
19 # Grouping anagrams
20 from collections import defaultdict
21 def group_anagrams(strs):
22     groups = defaultdict(list)
23     for s in strs:
24         groups[tuple(sorted(s))].append(s)
25     return list(groups.values())

```

Interview takeaway: Hash map is your first tool when you need $O(1)$ lookup. "Two pointer $O(n^2)$? Can we trade space for time?" → hash map. Watch out: dict keys must be hashable (no lists as keys; use tuples).

Real interview use-case: Two Sum, subarray sum equals k, longest consecutive sequence, top K frequent elements, valid sudoku.

3.4 Linked Lists

What it is: Sequence of nodes where each node holds a value and a pointer to the next (singly) or both next and previous (doubly) nodes. Non-contiguous memory.

Internal mechanics:

```

Singly Linked List:
  head
  ↓
  [1|·]→[2|·]→[3|·]→[4|·]→None
  node  next pointer

Doubly Linked List:
  head                                tail
  ↓                                  ↓
None←[·|1|·]↔[·|2|·]↔[·|3|·]↔[·|4|·]→None
      prev val next

Insert at head (O(1)):
  new_node.next = head
  head = new_node

Insert at tail (O(1) with tail pointer):
  tail.next = new_node
  tail = new_node

Delete node (given reference, doubly linked, O(1)):
  node.prev.next = node.next
  node.next.prev = node.prev

```


Delete node (singly linked, given reference only, $O(n)$):
 Must traverse from head to find predecessor
 → trick: copy next node's value into current, delete next

Complexity:

Op	Singly LL	Doubly LL
Access by index	$O(n)$	$O(n)$
Search	$O(n)$	$O(n)$
Insert at head	$O(1)$	$O(1)$
Insert at tail	$O(n)/O(1)$	$O(1)$
Insert at middle	$O(n)$	$O(n)$
Delete at head	$O(1)$	$O(1)$
Delete by ref	$O(n)$	$O(1)$

When to use:

- › Frequent insertions/deletions at head or known positions
- › Implementing stacks, queues, LRU cache
- › When you don't need random access

When NOT to use:

- › Random access needed (use array)
- › Memory is tight (each node has pointer overhead)
- › Cache performance critical (non-contiguous = cache misses)

Key Python patterns:

```

1 class ListNode:
2     def __init__(self, val=0, next=None):
3         self.val = val
4         self.next = next
5
6 # Reverse a linked list (iterative)
7 def reverse(head):
8     prev = None
9     curr = head
10    while curr:
11        nxt = curr.next
12        curr.next = prev
13        prev = curr
14        curr = nxt
15    return prev
16
17 # Floyd's cycle detection
18 def has_cycle(head):
19     slow = fast = head
20     while fast and fast.next:
21         slow = slow.next
22         fast = fast.next.next
23     if slow is fast:
24         return True
25     return False
26
27 # Find middle (slow/fast pointers)

```

```

28 def find_middle(head):
29     slow = fast = head
30     while fast and fast.next:
31         slow = slow.next
32         fast = fast.next.next
33     return slow
34
35 # Dummy node pattern (simplifies edge cases)
36 def delete_nth_from_end(head, n):
37     dummy = ListNode(0, head)
38     fast = slow = dummy
39     for _ in range(n + 1):
40         fast = fast.next
41     while fast:
42         fast = fast.next
43         slow = slow.next
44     slow.next = slow.next.next
45     return dummy.next

```

Interview takeaway: Always use a dummy head node to avoid null checks at head. Slow/fast pointer solves cycle detection, middle finding, and nth-from-end in one pass. Linked list problems are usually just pointer manipulation – draw it out.

Real interview use-case: Reverse linked list, merge two sorted lists, detect cycle, reorder list, LRU cache.

3.5 Stacks & Queues

What it is:

- › **Stack:** LIFO (Last In, First Out). Like a stack of plates.
- › **Queue:** FIFO (First In, First Out). Like a line at a store.
- › **Deque:** Double-ended queue; insert/delete at both ends $O(1)$.
- › **Monotonic stack/queue:** Stack/queue that maintains monotone order for range queries.
- › **Circular buffer:** Fixed-size ring buffer for queue; avoids shifting.

Internal mechanics:

```

Stack (LIFO):
Push 1,2,3: [1]→[2]→[3]  ← top is 3
Pop:       [1]→[2]       ← returns 3

Queue (FIFO) using circular buffer:
Capacity=5, head=0, tail=0
Enqueue 1,2,3:
[1][2][3][ ][ ]  head=0, tail=3
Dequeue:
[ ][2][3][ ][ ]  head=1, tail=3
Enqueue 4,5:
[ ][2][3][4][5]  head=1, tail=0 (wrapped!)
→ indices mod capacity: (tail+1) % cap

Monotonic Stack (next greater element):
arr = [2, 1, 5, 3, 4]
Stack tracks indices of elements waiting for their "next greater"
i=0: push 0      stack=[0]      (arr[0]=2)
i=1: 1<2, push 1  stack=[0,1]    (arr[1]=1)
i=2: 5>1, pop 1→ans[1]=5, 5>2 pop 0→ans[0]=5, push 2
      stack=[2]
i=3: 3<5, push 3  stack=[2,3]
i=4: 4>3, pop 3→ans[3]=4, 4<5 push 4
      stack=[2,4]

```

Remaining: no greater element $\rightarrow -1$
 ans = [5, 5, -1, 4, -1]

Deque (collections.deque in Python):
 appendleft / popleft $\rightarrow O(1)$
 append / pop $\rightarrow O(1)$

Complexity:

Op	Stack	Queue	Deque	Monotonic Stack
Push/Enqueue	$O(1)$	$O(1)$	$O(1)$	$O(1)$ amort
Pop/Dequeue	$O(1)$	$O(1)$	$O(1)$	$O(1)$ amort
Peek (top/front)	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Search	$O(n)$	$O(n)$	$O(n)$	$O(n)$

When to use:

- › Stack: DFS, expression parsing, undo operations, backtracking
- › Queue: BFS, task scheduling, sliding window with deque
- › Monotonic stack: next greater/smaller element, largest rectangle in histogram
- › Deque: sliding window maximum, palindrome check

Key Python patterns:

```

1 from collections import deque
2
3 # Stack (use list, pop from end)
4 stack = []
5 stack.append(x)    # push
6 stack.pop()        # pop  $O(1)$ 
7 stack[-1]          # peek
8
9 # Queue (use deque for  $O(1)$  dequeue)
10 q = deque()
11 q.append(x)         # enqueue
12 q.popleft()         # dequeue  $O(1)$ 
13
14 # BFS template
15 def bfs(root):
16     q = deque([root])
17     while q:
18         node = q.popleft()
19         for child in node.children:
20             q.append(child)
21
22 # Monotonic stack - next greater element
23 def next_greater(nums):
24     n = len(nums)
25     ans = [-1] * n
26     stack = [] # indices
27     for i, num in enumerate(nums):
28         while stack and nums[stack[-1]] < num:
29             idx = stack.pop()
30             ans[idx] = num
31         stack.append(i)
32     return ans
33

```

```

34 # Sliding window maximum (monotonic deque)
35 def sliding_max(nums, k):
36     dq = deque() # indices, decreasing values
37     res = []
38     for i, num in enumerate(nums):
39         while dq and nums[dq[-1]] < num:
40             dq.pop()
41         dq.append(i)
42         if dq[0] == i - k: # out of window
43             dq.popleft()
44         if i ≥ k - 1:
45             res.append(nums[dq[0]])
46     return res

```

Interview takeaway: Monotonic stack is the pattern for "next greater/smaller element" and "largest rectangle" problems. BFS always uses a queue. For sliding window max/min, use a monotonic deque. Never use `list.pop(0)` for queues — it's $O(n)$; use `deque.popleft()`.

Real interview use-case: Valid parentheses, min stack, daily temperatures, largest rectangle in histogram, sliding window maximum.

3.6 Trees

What it is: Hierarchical, acyclic connected graph. Root at top, leaves at bottom.

Internal mechanics:

Binary Tree (each node: up to 2 children):



BST (Binary Search Tree): left < root < right



BST invariant: for any node N:

- All nodes in left subtree < N.val
- All nodes in right subtree > N.val
- In-order traversal gives sorted sequence!

BST Search:

Find 6: start at 8 → 6 < 8 go left → 3 → 6 > 3 go right → 6 ✓
 $O(h)$ where h = height. Balanced: $h = \log n$. Skewed: $h = n$.

Balanced Tree (AVL) — height constraint:

Balance factor = height(left) - height(right)
 Must be in $\{-1, 0, 1\}$ for every node

Rotation on violation:

- Left-Left: right rotate
- Right-Right: left rotate
- Left-Right: left rotate then right rotate
- Right-Left: right rotate then left rotate

Red-Black Tree (less strict, used in Java TreeMap):

Properties:

1. Every node is red or black
2. Root is black
3. Red nodes have black children

4. All paths from node to null have same # black nodes
 → Height $\leq 2 \cdot \log(n+1)$ guaranteed

Tree Traversals:

Pre-order: root → left → right [1,2,4,5,3,6]
 In-order: left → root → right [4,2,5,1,3,6] ← sorted for BST!
 Post-order: left → right → root [4,5,2,6,3,1]
 Level-order: BFS level by level [1,2,3,4,5,6]

Complexity:

Op	Balanced BST	Unbalanced BST	AVL / RB Tree
Search	$O(\log n)$	$O(n)$	$O(\log n)$
Insert	$O(\log n)$	$O(n)$	$O(\log n)$
Delete	$O(\log n)$	$O(n)$	$O(\log n)$
Min/Max	$O(\log n)$	$O(n)$	$O(\log n)$
In-order	$O(n)$	$O(n)$	$O(n)$

When to use:

- › BST: sorted data with dynamic insertions/deletions
- › AVL/RB: guaranteed $O(\log n)$ when data can be adversarial
- › Any tree: hierarchical relationships, recursive decomposition

Key Python patterns:

```

1 class TreeNode:
2     def __init__(self, val=0, left=None, right=None):
3         self.val = val
4         self.left = left
5         self.right = right
6
7 # In-order traversal (recursive)
8 def inorder(root):
9     if not root: return []
10    return inorder(root.left) + [root.val] + inorder(root.right)
11
12 # In-order traversal (iterative - important for interviews)
13 def inorder_iter(root):
14     result, stack = [], []
15     curr = root
16     while curr or stack:
17         while curr:
18             stack.append(curr)
19             curr = curr.left
20         curr = stack.pop()
21         result.append(curr.val)
22         curr = curr.right
23     return result
24
25 # Height of tree
26 def height(root):
27     if not root: return 0
28     return 1 + max(height(root.left), height(root.right))
29
30 # Check if BST is valid
31 def is_valid_bst(root, lo=float('-inf'), hi=float('inf')):
```

```

32     if not root: return True
33     if not (lo < root.val < hi): return False
34     return (is_valid_bst(root.left, lo, root.val) and
35             is_valid_bst(root.right, root.val, hi))
36
37 # LCA (Lowest Common Ancestor) in BST
38 def lca_bst(root, p, q):
39     if p.val < root.val and q.val < root.val:
40         return lca_bst(root.left, p, q)
41     if p.val > root.val and q.val > root.val:
42         return lca_bst(root.right, p, q)
43     return root
44
45 # Level order traversal
46 from collections import deque
47 def level_order(root):
48     if not root: return []
49     result, q = [], deque([root])
50     while q:
51         level = []
52         for _ in range(len(q)):
53             node = q.popleft()
54             level.append(node.val)
55             if node.left: q.append(node.left)
56             if node.right: q.append(node.right)
57         result.append(level)
58     return result

```

Interview takeaway: BST in-order = sorted. Use recursion with min/max bounds for BST validation, not just $\text{left} < \text{root} < \text{right}$ (common bug). Diameter of tree = longest path through any node = $\max(\text{left_depth} + \text{right_depth})$ over all nodes.

Real interview use-case: Validate BST, lowest common ancestor, diameter of binary tree, serialize/deserialize tree, path sum, kth smallest in BST.

3.7 Heaps / Priority Queues

What it is: A complete binary tree stored as an array, where each parent satisfies the heap property (min-heap: $\text{parent} \leq \text{children}$; max-heap: $\text{parent} \geq \text{children}$). Top element is always the min (or max).

Internal mechanics:

Min-Heap (array representation):



Array: [1, 3, 5, 7, 4, 8, 9]

Index: 0 1 2 3 4 5 6

Index relationships:

Parent(i) = (i-1) // 2

Left child(i) = 2*i + 1

Right child(i) = 2*i + 2

Sift Up (after insert at end):

Insert 2: append → [1,3,5,7,4,8,9,2]

Compare 2 with parent (index 3): arr[3]=7 > 2 → swap
→ [1,3,5,2,4,8,9,7]

Compare 2 with parent (index 1): arr[1]=3 > 2 → swap

→ [1,2,5,3,4,8,9,7]
 Compare 2 with parent (index 0): $\text{arr}[0]=1 < 2 \rightarrow \text{stop} \checkmark$

Sift Down (after extracting root – put last element at root):

Extract 1: put 7 at root → [7,2,5,3,4,8,9]
 Compare 7 with children 2,5: min child=2 → swap
 → [2,7,5,3,4,8,9]
 Compare 7 with children 3,4: min child=3 → swap
 → [2,3,5,7,4,8,9]
 7 has no children → stop $\checkmark$

Build Heap (heapify) from arbitrary array – $O(n)$:

Start sifting down from last non-leaf ($n//2 - 1$) to root
 Counterintuitive: $O(n)$ not $O(n \log n)$ because most nodes are near leaves

Complexity:

Op	Time
Get min/max	$O(1)$
Insert	$O(\log n)$
Extract min	$O(\log n)$
Build heap	$O(n)$
Search	$O(n)$
Heapsort	$O(n \log n)$

When to use:

- › Get the min/max repeatedly (priority queues)
- › Top K elements (use heap of size K)
- › Merge K sorted lists
- › Dijkstra's shortest path
- › Median maintenance (two heaps)

Key Python patterns:

```

1 import heapq
2
3 # Min-heap (default)
4 heap = []
5 heapq.heappush(heap, 3)
6 heapq.heappush(heap, 1)
7 heapq.heappush(heap, 2)
8 heapq.heappop(heap)    # returns 1
9
10 # Max-heap: negate values
11 max_heap = []
12 heapq.heappush(max_heap, -3)
13 heapq.heappush(max_heap, -1)
14 -heapq.heappop(max_heap)    # returns 3
15
16 # Build heap in O(n)
17 nums = [3, 1, 4, 1, 5, 9]
18 heapq.heapify(nums)
19
20 # Top K elements
21 def top_k(nums, k):
22     return heapq.nlargest(k, nums)    # O(n log k)
  
```

```

23
24 # Top K using min-heap of size k (space efficient)
25 def top_k_heap(nums, k):
26     heap = []
27     for n in nums:
28         heapq.heappush(heap, n)
29         if len(heap) > k:
30             heapq.heappop(heap)
31     return sorted(heap, reverse=True)
32
33 # Merge K sorted lists
34 def merge_k_lists(lists):
35     heap = []
36     for i, lst in enumerate(lists):
37         if lst:
38             heapq.heappush(heap, (lst[0], i, 0))
39     result = []
40     while heap:
41         val, i, j = heapq.heappop(heap)
42         result.append(val)
43         if j + 1 < len(lists[i]):
44             heapq.heappush(heap, (lists[i][j+1], i, j+1))
45     return result
46
47 # Median maintenance (two heaps)
48 class MedianFinder:
49     def __init__(self):
50         self.lo = [] # max-heap (negate)
51         self.hi = [] # min-heap
52     def add_num(self, num):
53         heapq.heappush(self.lo, -num)
54         heapq.heappush(self.hi, -heapq.heappop(self.lo))
55         if len(self.hi) > len(self.lo):
56             heapq.heappush(self.lo, -heapq.heappop(self.hi))
57     def find_median(self):
58         if len(self.lo) > len(self.hi): return -self.lo[0]
59         return (-self.lo[0] + self.hi[0]) / 2.0

```

Interview takeaway: Python only has min-heap via `heapq`; negate values for max-heap. `heapify` is $O(n)$. Top K smallest: use max-heap of size K and push out the max when size exceeds K. Top K largest: use min-heap of size K.

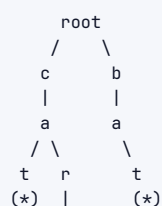
Real interview use-case: Kth largest element, merge K sorted lists, task scheduler, find median from data stream, Dijkstra's.

3.8 Tries (Prefix Trees)

What it is: A tree where each path from root to leaf spells a word, and each node represents one character. Shared prefixes share nodes.

Internal mechanics:

Trie storing: ["cat", "car", "card", "care", "bat"]




```

    / \
   d   e
  (*)  (*)

```

(*) = is_end marker

Node structure:

```

children: dict[char → TrieNode] (or array[26])
is_end: bool

```

Insert "card":

```

root → c (create if absent)
      → a (create if absent)
      → r (exists, reuse)
      → d (create)
mark d.is_end = True

```

Search "care":

```

root → c → a → r → e → is_end? True ✓

```

StartsWith "ca":

```

root → c → a → exists? True ✓ (prefix search)

```

Space: each node uses $O(\text{ALPHA})$ where $\text{ALPHA}=26$ for lowercase
 n strings of avg length m: $O(n*m)$ nodes worst case
 (better than storing all strings if many shared prefixes)

Complexity (m = length of key):

Op	Time	Space
Insert	$O(m)$	$O(m)$
Search	$O(m)$	$O(1)$
StartsWith	$O(m)$	$O(1)$
Delete	$O(m)$	$O(1)$

When to use:

- › Autocomplete / prefix search
- › Spell checking
- › IP routing tables
- › Word search in a grid

When NOT to use:

- › Simple key-value lookup (use hash map)
- › Keys are not strings or have no shared prefixes

Key Python patterns:

```

class TrieNode:
    def __init__(self):
        self.children = {}
        self.is_end = False

class Trie:
    def __init__(self):
        self.root = TrieNode()

    def insert(self, word: str) → None:
        node = self.root

```

```

12     for ch in word:
13         if ch not in node.children:
14             node.children[ch] = TrieNode()
15             node = node.children[ch]
16         node.is_end = True
17
18     def search(self, word: str) → bool:
19         node = self.root
20         for ch in word:
21             if ch not in node.children:
22                 return False
23             node = node.children[ch]
24         return node.is_end
25
26     def starts_with(self, prefix: str) → bool:
27         node = self.root
28         for ch in prefix:
29             if ch not in node.children:
30                 return False
31             node = node.children[ch]
32         return True
33
34 # Word search II (trie + DFS backtracking)
35 # Build trie from word list, then DFS on board
36 # When we reach a trie end node, we found a word

```

Interview takeaway: Trie is the answer to "find all words with prefix X" or "word search on a grid with a word list". For word search II (board + word list), build a trie from words and run DFS on the board — this prunes early when no prefix matches.

Real interview use-case: Implement trie, word search II, replace words, design search autocomplete, map sum pairs.

3.9 Graphs

What it is: A set of vertices (nodes) connected by edges. Can be directed/undirected, weighted/unweighted, cyclic/acyclic.

Internal mechanics:

Graph: 4 vertices, 4 edges (undirected)

```

0 --- 1
|  /  |
| /   |
| /   |
2 --- 3

```

Adjacency List (preferred for sparse graphs):

```

{
  0: [1, 2],
  1: [0, 2, 3],
  2: [0, 1, 3],
  3: [1, 2]
}

```

Space: $O(V + E)$

Check edge (u,v): $O(\text{degree}(u))$

Get neighbors: $O(\text{degree}(u))$

Adjacency Matrix (preferred for dense graphs):

```

    0  1  2  3
0 [ 0  1  1  0 ]
1 [ 1  0  1  1 ]
2 [ 1  1  0  1 ]

```

```

3 [ 0 1 1 0 ]
Space:  $O(V^2)$ 
Check edge (u,v):  $O(1)$ 
Get neighbors:  $O(V)$ 

Weighted adjacency list:
{0: [(1, 4), (2, 1)], ...} ← (neighbor, weight)

Types:
Undirected: edges go both ways
Directed (Digraph): edges have direction
DAG: Directed Acyclic Graph (used in topological sort)
Tree: connected undirected graph with  $V-1$  edges

```

Complexity:

Op	Adj List	Adj Matrix
Space	$O(V+E)$	$O(V^2)$
Add vertex	$O(1)$	$O(V^2)$
Add edge	$O(1)$	$O(1)$
Remove edge	$O(E)$	$O(1)$
Check edge (u,v)	$O(\text{degree})$	$O(1)$
Get all neighbors	$O(\text{degree})$	$O(V)$
BFS/DFS	$O(V+E)$	$O(V^2)$
Dijkstra	$O((V+E)\log V)$	$O(V^2\log V)$

When to use:

- › Adj List: default, sparse graphs, BFS/DFS heavy
- › Adj Matrix: dense graphs, need $O(1)$ edge existence check, Floyd-Warshall

Key Python patterns:

```

1 from collections import defaultdict, deque
2
3 # Build graph
4 graph = defaultdict(list)
5 for u, v in edges:
6     graph[u].append(v)
7     graph[v].append(u) # undirected
8
9 # BFS (shortest path in unweighted graph)
10 def bfs(graph, start, end):
11     visited = {start}
12     q = deque([(start, 0)])
13     while q:
14         node, dist = q.popleft()
15         if node == end: return dist
16         for nei in graph[node]:
17             if nei not in visited:
18                 visited.add(nei)
19                 q.append((nei, dist + 1))
20     return -1
21
22 # DFS (iterative)
23 def dfs(graph, start):
24     visited = set()

```

```

25     stack = [start]
26     while stack:
27         node = stack.pop()
28         if node in visited: continue
29         visited.add(node)
30         for nei in graph[node]:
31             stack.append(nei)
32
33 # DFS (recursive)
34 def dfs_rec(node, visited, graph):
35     visited.add(node)
36     for nei in graph[node]:
37         if nei not in visited:
38             dfs_rec(nei, visited, graph)
39
40 # Topological sort (Kahn's algorithm, BFS-based)
41 def topo_sort(n, edges):
42     graph = defaultdict(list)
43     in_degree = [0] * n
44     for u, v in edges:
45         graph[u].append(v)
46         in_degree[v] += 1
47     q = deque(i for i in range(n) if in_degree[i] == 0)
48     order = []
49     while q:
50         node = q.popleft()
51         order.append(node)
52         for nei in graph[node]:
53             in_degree[nei] -= 1
54             if in_degree[nei] == 0:
55                 q.append(nei)
56     return order if len(order) == n else [] # empty = cycle exists
57
58 # Dijkstra's (single source shortest path)
59 import heapq
60 def dijkstra(graph, start):
61     dist = {node: float('inf') for node in graph}
62     dist[start] = 0
63     heap = [(0, start)]
64     while heap:
65         d, node = heapq.heappop(heap)
66         if d > dist[node]: continue
67         for nei, weight in graph[node]:
68             new_d = d + weight
69             if new_d < dist[nei]:
70                 dist[nei] = new_d
71                 heapq.heappush(heap, (new_d, nei))
72     return dist

```

Interview takeaway: BFS for shortest path in unweighted graph. Dijkstra for weighted (non-negative). Detect cycle in directed graph: DFS with in-stack tracking. Topological sort only valid for DAGs. "Number of islands" = number of DFS/BFS calls on unvisited cells.

Real interview use-case: Number of islands, clone graph, course schedule, network delay time, Pacific Atlantic water flow, word ladder.

3.10 Union-Find / Disjoint Set

What it is: Data structure to track which elements belong to the same connected component. Supports two operations: find (which component?) and union (merge components).

Internal mechanics:

```
Initial state (6 elements, each its own component):
parent = [0, 1, 2, 3, 4, 5]
rank   = [0, 0, 0, 0, 0, 0]

Union(0, 1): find roots → 0,1; rank equal → make 1 child of 0
parent = [0, 0, 2, 3, 4, 5]
rank   = [1, 0, 0, 0, 0, 0]

Union(2, 3):
parent = [0, 0, 2, 2, 4, 5]

Union(0, 2): find roots → 0, 2; rank[0]=1 > rank[2]=0 → 2 under 0
parent = [0, 0, 0, 2, 4, 5] ← 2's parent is now 0

Path Compression during find(3):
find(3): parent[3]=2, parent[2]=0, parent[0]=0 → root=0
Compress: set parent[3]=0, parent[2]=0 directly
parent = [0, 0, 0, 0, 4, 5] ← next find(3) is O(1)

After path compression and union by rank:
find and union are effectively  $O(\alpha(n)) \approx O(1)$ 
where  $\alpha$  = inverse Ackermann function
 $\alpha(n) < 5$  for any practical  $n$ 

Visual after several unions:
Component 1: {0,1,2,3} ← tree rooted at 0
Component 2: {4}
Component 3: {5}
    0
   /\
  1 2 3 ← after path compression, flat tree
```

Complexity:

Op	Naive	Union by Rank	+ Path Compression
Find	$O(n)$	$O(\log n)$	$O(\alpha(n)) \approx O(1)$
Union	$O(n)$	$O(\log n)$	$O(\alpha(n)) \approx O(1)$
Space	$O(n)$	$O(n)$	$O(n)$

When to use:

- › Connected components (dynamic)
- › Detect cycles in undirected graph
- › Kruskal's minimum spanning tree
- › Grouping problems ("same group as")

Key Python patterns:

```
Python
1 class UnionFind:
2     def __init__(self, n):
3         self.parent = list(range(n))
4         self.rank = [0] * n
5         self.components = n
6
7     def find(self, x):
8         if self.parent[x] != x:
9             self.parent[x] = self.find(self.parent[x]) # path compression
10        return self.parent[x]
11
```

```

12     def union(self, x, y):
13         rx, ry = self.find(x), self.find(y)
14         if rx == ry: return False # already connected
15         # Union by rank
16         if self.rank[rx] < self.rank[ry]:
17             rx, ry = ry, rx
18         self.parent[ry] = rx
19         if self.rank[rx] == self.rank[ry]:
20             self.rank[rx] += 1
21         self.components -= 1
22         return True
23
24     def connected(self, x, y):
25         return self.find(x) == self.find(y)
26
27 # Example: count components in graph
28 def count_components(n, edges):
29     uf = UnionFind(n)
30     for u, v in edges:
31         uf.union(u, v)
32     return uf.components
33
34 # Detect cycle in undirected graph
35 def has_cycle(n, edges):
36     uf = UnionFind(n)
37     for u, v in edges:
38         if not uf.union(u, v): # already in same component
39             return True
40     return False
41
42 # Kruskal's MST
43 def kruskal(n, edges):
44     uf = UnionFind(n)
45     edges.sort(key=lambda e: e[2]) # sort by weight
46     mst_cost = 0
47     for u, v, w in edges:
48         if uf.union(u, v):
49             mst_cost += w
50     return mst_cost

```

Interview takeaway: Always implement both path compression AND union by rank. Without both, you don't get the near- $O(1)$ guarantee. Union-Find cannot efficiently handle edge deletion — if you need that, use Link-Cut Trees (not interview scope).

Real interview use-case: Number of provinces, redundant connection, accounts merge, satisfiability of equality equations.

3.11 LRU / LFU Cache Structures

What it is:

- › **LRU (Least Recently Used):** Evicts the item not accessed for the longest time. $O(1)$ get and put.
- › **LFU (Least Frequently Used):** Evicts the item used fewest times (break ties by LRU). $O(1)$ all operations.

Internal mechanics:

LRU Cache (capacity=3):
 Internals: HashMap + Doubly Linked List
 Map: key → node (O(1) access)
 DLL: most recent at front, least recent at tail

get(1): move node 1 to front
 put(4): if full, remove tail (LRU), insert at front, update map

State after put(1), put(2), put(3):
 [3] ↔ [2] ↔ [1] ← tail (LRU)
 ↑
 head (MRU)

get(1): move 1 to head
 [1] ↔ [3] ↔ [2] ← tail (LRU)

put(4): evict tail (2), add 4 at head
 [4] ↔ [1] ↔ [3]

LFU Cache (capacity=3):
 Internals: HashMap(key→(val,freq)) + HashMap(freq→DLL of keys)
 + min_freq tracker
 On access: increment freq, move to freq+1 list
 On evict: remove LRU from min_freq list
 Trickier: when min_freq list empties, min_freq++

Complexity:

Op	LRU	LFU
get	O(1)	O(1)
put	O(1)	O(1)

Key Python patterns:

```

1 from collections import OrderedDict
2
3 # LRU using OrderedDict (move_to_end trick)
4 class LRUCache:
5     def __init__(self, capacity: int):
6         self.cap = capacity
7         self.cache = OrderedDict() # key → value, ordered by access time
8
9     def get(self, key: int) → int:
10        if key not in self.cache: return -1
11        self.cache.move_to_end(key) # mark as most recently used
12        return self.cache[key]
13
14    def put(self, key: int, value: int) → None:
15        if key in self.cache:
16            self.cache.move_to_end(key)
17        self.cache[key] = value
18        if len(self.cache) > self.cap:
19            self.cache.popitem(last=False) # remove LRU (first item)
20
21 # LRU from scratch (HashMap + DLL) – what interviewers actually want
22 class DLLNode:
23     def __init__(self, key=0, val=0):
24         self.key = key
25         self.val = val
26         self.prev = self.next = None
27

```

```

28 class LRUCacheManual:
29     def __init__(self, capacity):
30         self.cap = capacity
31         self.cache = {}
32         self.head = DLLNode() # dummy head (MRU side)
33         self.tail = DLLNode() # dummy tail (LRU side)
34         self.head.next = self.tail
35         self.tail.prev = self.head
36
37     def _remove(self, node):
38         node.prev.next = node.next
39         node.next.prev = node.prev
40
41     def _insert_front(self, node):
42         node.next = self.head.next
43         node.prev = self.head
44         self.head.next.prev = node
45         self.head.next = node
46
47     def get(self, key):
48         if key not in self.cache: return -1
49         node = self.cache[key]
50         self._remove(node)
51         self._insert_front(node)
52         return node.val
53
54     def put(self, key, val):
55         if key in self.cache:
56             self._remove(self.cache[key])
57         node = DLLNode(key, val)
58         self.cache[key] = node
59         self._insert_front(node)
60         if len(self.cache) > self.cap:
61             lru = self.tail.prev
62             self._remove(lru)
63             del self.cache[lru.key]

```

Interview takeaway: LRU = HashMap + DLL with dummy head and tail. The dummy nodes eliminate all edge-case null checks. LFU is significantly harder – use HashMap(key→freq) + HashMap(freq→OrderedDict) + min_freq variable.

Real interview use-case: LRU cache (literal LeetCode 146), LFU cache (LeetCode 460), design Twitter, design browser history.

3.12 Segment Tree & Fenwick Tree (BIT)

What it is:

- ▶ **Segment Tree:** Tree where each node stores aggregate (sum/min/max) over a range. Supports range queries and point/range updates in $O(\log n)$.
- ▶ **Fenwick Tree (Binary Indexed Tree):** Array-based structure using binary representation for prefix sums. Simpler to implement; less general.

Internal mechanics:

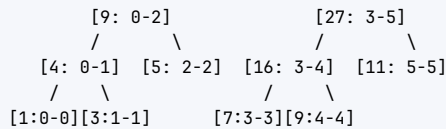
Array: [1, 3, 5, 7, 9, 11] (0-indexed)

Segment Tree (sum):

```

    [36: 0-5]
    /      \

```

Query $\text{sum}(1,4)$: decompose into $[1-2]$ and $[3-4] \rightarrow 8 + 16 = 24$
 Update $\text{arr}[2] = 10$: update leaf, propagate to root $\rightarrow O(\log n)$
 Space: $O(4n)$ for array representation

Fenwick Tree (BIT) for prefix sums:

Index: 1 2 3 4 5 6
 arr: [1, 3, 5, 7, 9, 11]
 BIT: [1, 4, 5, 16, 9, 20] ← each stores sum of specific range

$\text{BIT}[i] = \text{sum of arr}[i - \text{lowbit}(i) + 1 \dots i]$
 $\text{lowbit}(i) = i \& (-i)$ ← isolates least significant bit

Prefix sum query(5):
 $i=5$ (101): add $\text{BIT}[5]=9$, $i=5-(5\&5)=4$
 $i=4$ (100): add $\text{BIT}[4]=16$, $i=4-(4\&4)=0 \rightarrow \text{stop}$
 Total: $9+16=25$ ✓ (but wait: $1+3+5+7+9=25$ ✓)

Update(3, delta=2): propagate up
 $i=3$: $\text{BIT}[3] += 2$, $i=3+(3\&3)=4$
 $i=4$: $\text{BIT}[4] += 2$, $i=4+(4\&4)=8 > n \rightarrow \text{stop}$

Complexity:

Op	Naive Array	Segment Tree	Fenwick Tree
Build	$O(n)$	$O(n)$	$O(n \log n)$
Point update	$O(1)$	$O(\log n)$	$O(\log n)$
Range update	$O(n)$	$O(\log n)$	$O(\log n)$
Range query	$O(n)$	$O(\log n)$	$O(\log n)$
Space	$O(n)$	$O(4n)$	$O(n)$

When to use:

- › Segment tree: range min/max queries, lazy propagation for range updates
- › Fenwick tree: only prefix sums needed; cleaner code
- › Both: dynamic range queries where array is mutated

Key Python patterns:

```

1 # Fenwick Tree (BIT) - prefix sum
2 class BIT:
3     def __init__(self, n):
4         self.n = n
5         self.tree = [0] * (n + 1) # 1-indexed
6
7     def update(self, i, delta): # add delta to position i
8         while i <= self.n:
9             self.tree[i] += delta
10            i += i & (-i) # move to parent
11
12    def query(self, i): # prefix sum [1..i]
13        s = 0
14        while i > 0:
15            s += self.tree[i]
16            i -= i & (-i) # move to ancestor

```

```

17         return s
18
19     def range_query(self, l, r): # sum [l..r]
20         return self.query(r) - self.query(l - 1)
21
22 # Segment Tree (iterative, cleaner)
23 class SegmentTree:
24     def __init__(self, nums):
25         self.n = len(nums)
26         self.tree = [0] * (2 * self.n)
27         # Build
28         for i, v in enumerate(nums):
29             self.tree[self.n + i] = v
30         for i in range(self.n - 1, 0, -1):
31             self.tree[i] = self.tree[2*i] + self.tree[2*i+1]
32
33     def update(self, i, val):
34         i += self.n
35         self.tree[i] = val
36         while i > 1:
37             i //= 2
38             self.tree[i] = self.tree[2*i] + self.tree[2*i+1]
39
40     def query(self, l, r): # sum [l, r) half-open
41         res = 0
42         l += self.n; r += self.n
43         while l < r:
44             if l & 1: res += self.tree[l]; l += 1
45             if r & 1: r -= 1; res += self.tree[r]
46             l //= 2; r //= 2
47         return res

```

Interview takeaway: Fenwick tree is simpler to code and sufficient for prefix sum + point update problems. Use segment tree when you need range updates or range min/max. The "count of smaller numbers after self" problem → BIT is the elegant solution.

Real interview use-case: Range sum query (mutable), count of smaller numbers after self, reverse pairs, falling squares.

3.13 Bloom Filter

What it is: A probabilistic, space-efficient data structure that answers "is X possibly in the set?" or "is X definitely NOT in the set?" — with false positives but no false negatives.

Internal mechanics:

Bloom Filter (m=20 bits, k=3 hash functions):

Initial state:

[0]

Insert "cat" (hash1=3, hash2=7, hash3=15):

[0][0][0][1][0][0][0][1][0][0][0][0][0][0][0][1][0][0][0][0]

 ↑ ↑ ↑

Insert "dog" (hash1=3, hash2=10, hash3=17):

[0][0][0][1][0][0][0][1][0][0][0][0][0][1][0][1][0][1][0][0]

Note: bit 3 now set by both "cat" and "dog"!

Query "cat": check bits 3,7,15 → all 1 → POSSIBLY in set ✓

Query "hat": hash1=3, hash2=10, hash3=17 → all 1 → POSSIBLY in set × FALSE POSITIVE!

Query "xyz": hash1=1 → 0 → DEFINITELY NOT in set ✓

False positive probability $\approx (1 - e^{(-kn/m)})^k$
 where n =items inserted, m =bit array size, k =hash functions

Key: no false negatives (if bloom says NO, it's definitely NO)
 cannot delete items (counter-based bloom filters can)
 cannot enumerate stored elements

Complexity:

Op	Time	Space
Insert	$O(k)$	$O(m)$
Query	$O(k)$	$O(1)$
Delete	N/A	-

When to use:

- › Cache pre-filtering: "has this URL been seen before?" before hitting DB
- › Distributed systems: "does key exist on this node?" before network call
- › Spell checkers: fast "not a word" filter
- › HBase/Cassandra: skip SSTable reads for non-existent keys

Ties to systems:

- › Cassandra uses bloom filters to avoid unnecessary disk I/O
- › Google Chrome used bloom filter for Safe Browsing URL blacklist
- › Bitcoin uses bloom filters in SPV (Simplified Payment Verification)

Interview takeaway: Bloom filters trade accuracy for space and speed. False positives OK; false negatives are impossible. When an interviewer asks "how would you check if a URL was seen before using $O(1)$ space?" – bloom filter is the answer. Note: you can NOT delete from a standard bloom filter.

4 Choosing the Right Data Structure

Decision table: "I need X $\rightarrow$ use Y"

Need	Use	Why
$O(1)$ access by index	Array / List	Contiguous memory
$O(1)$ key-value lookup	Hash Map	Hash function
Ordered key-value, range queries	BST / Sorted Map	In-order traversal
Get min/max repeatedly	Heap	Root always min/max
Get kth smallest/largest	Heap of size k	$O(n \log k)$
Prefix search / autocomplete	Trie	Shared prefix compression
LIFO (undo, recursion)	Stack	Push/pop from one end
FIFO (BFS, scheduling)	Queue / Deque	Enqueue/dequeue $O(1)$
Sliding window max/min	Monotonic Deque	Maintain monotone order
Next greater/smaller element	Monotonic Stack	Classic pattern
Connected components	Union-Find	Near- $O(1)$ union/find
Shortest path (unweighted)	BFS (Queue)	Level-by-level expansion
Shortest path (weighted, non-neg)	Dijkstra (Heap)	Greedy + priority queue

Need	Use	Why
Shortest path (negative weights)	Bellman-Ford	DP approach
All pairs shortest path	Floyd-Warshall	$O(V^3)$ DP
Range sum query + updates	Fenwick Tree / Segment Tree	$O(\log n)$ both
Range min/max query + updates	Segment Tree	BIT can't do min/max
$O(1)$ LRU eviction	HashMap + DLL	$O(1)$ access + $O(1)$ order
Membership test, space-constrained	Bloom Filter	Probabilistic, tiny space
Count frequencies	HashMap / Counter	Hash map with integer values
Find duplicates	HashSet or sorting	$O(n)$ with hash, $O(n \log n)$ sort
Merge k sorted sequences	Min-Heap of k elements	$O(n \log k)$
Detect cycle (undirected graph)	Union-Find	Simpler than DFS coloring
Detect cycle (directed graph)	DFS with in-stack tracking	Union-Find doesn't work here
Topological ordering	Kahn's (BFS) or DFS	Works on DAGs only

5 Language Notes

5.1 Python

Need	Python	Notes
Dynamic array	list	append $O(1)$ amortized
Hash map	dict	defaultdict(list/int/set)
Hash set	set	add, in $O(1)$
Ordered map (by key)	sortedcontainers.SortedList	Not stdlib; use bisect + list
Min heap	heapq (module)	Negate for max heap
Queue / Deque	collections.deque	appendleft, popleft $O(1)$
Counter	collections.Counter	most_common(k)
LRU built-in	collections.OrderedDict	move_to_end, popitem
Bisect (sorted insert)	bisect.bisect_left/right	Binary search on sorted list
Infinity	float('inf')	
Char to int	ord('a') $\rightarrow$ 97	chr(97) $\rightarrow$ 'a'

```

1 # Common Python idioms
2 from collections import defaultdict, Counter, deque, OrderedDict
3 import heapq
4 from bisect import bisect_left, bisect_right, insort
5
6 # Heap with custom key (push tuples)
7 heapq.heappush(heap, (priority, item))
8
9 # Sorted list simulation with bisect
10 import bisect
11 sorted_list = []
12 bisect.insort(sorted_list, 5) # insert and maintain order
13 idx = bisect.bisect_left(sorted_list, 5) # binary search

```

```

14
15 # defaultdict prevents KeyError
16 graph = defaultdict(list)
17 count = defaultdict(int)

```

5.2 Java

Need	Java Class	Notes
Dynamic array	<code>ArrayList<T></code>	<code>add()</code> $O(1)$ amortized
Hash map	<code>HashMap<K,V></code>	Not ordered
Ordered map (by key)	<code>TreeMap<K,V></code>	Red-Black tree, $O(\log n)$ ops
Ordered set	<code>TreeSet<T></code>	Red-Black tree
Hash set	<code>HashSet<T></code>	
Linked hash map	<code>LinkedHashMap<K,V></code>	Insertion-order iteration (LRU)
Min/Max heap	<code>PriorityQueue<T></code>	Min by default; pass <code>Comparator.reverseOrder()</code> for max
Stack	<code>Deque<T></code> as stack	Prefer <code>ArrayDeque</code> over Stack
Queue	<code>ArrayDeque<T></code>	Or <code>LinkedList<T></code>
Deque	<code>ArrayDeque<T></code>	

```

1 // Java idioms
2 PriorityQueue<Integer> minHeap = new PriorityQueue<>();
3 PriorityQueue<Integer> maxHeap = new PriorityQueue<>(Collections.reverseOrder());
4
5 Map<String, Integer> map = new HashMap<>();
6 map.getOrDefault(key, 0);
7 map.computeIfAbsent(key, k -> new ArrayList<>()).add(val);
8
9 Deque<Integer> stack = new ArrayDeque<>(); // use as stack
10 stack.push(1); stack.pop(); stack.peek();
11
12 Queue<Integer> queue = new ArrayDeque<>(); // use as queue
13 queue.offer(1); queue.poll(); queue.peek();
14
15 TreeMap<Integer, Integer> ordered = new TreeMap<>();
16 ordered.floorKey(x); // largest key ≤ x
17 ordered.ceilingKey(x); // smallest key ≥ x

```

5.3 C++

Need	C++ Container	Notes
Dynamic array	<code>vector<T></code>	<code>push_back</code> $O(1)$ amortized
Hash map	<code>unordered_map<K,V></code>	$O(1)$ avg
Ordered map (by key)	<code>map<K,V></code>	Red-Black tree, $O(\log n)$
Hash set	<code>unordered_set<T></code>	
Ordered set	<code>set<T></code>	+ <code>lower_bound</code> , <code>upper_bound</code>
Priority queue (max)	<code>priority_queue<T></code>	Max by default

Need	C++ Container	Notes
Priority queue (min)	<code>priority_queue<T, vector<T>, greater<T>></code>	
Stack	<code>stack<T></code>	
Queue	<code>queue<T></code>	
Deque	<code>deque<T></code>	

```

1 // C++ idioms
2 #include <bits/stdc++.h>
3 using namespace std;
4
5 unordered_map<int,int> freq;
6 freq[x]++; // default-initializes to 0
7
8 // Min heap
9 priority_queue<int, vector<int>, greater<int>> minHeap;
10
11 // Ordered set with lower_bound
12 set<int> s;
13 auto it = s.lower_bound(x); // O(log n)
14
15 // Sort with custom comparator
16 sort(v.begin(), v.end(), [](auto& a, auto& b){ return a[0] < b[0]; });

```

6 Real-Life Analogies

Data Structure	Analogy
Array	Numbered mailboxes in a row — access box 42 instantly
Hash Map	Library card catalog — look up by title, get shelf location
Linked List	Train cars — each car knows only the next car
Stack	Stack of plates — only access the top
Queue	Queue at a bank — first in, first served
Deque	Train that can board/exit from both ends
BST	Phone book — organized so you can binary search
Heap	Priority ER room — most critical patient always treated first
Trie	Autocomplete on your phone — shared prefix saves space
Graph	Road map — cities are nodes, roads are edges
Union-Find	Social groups — merging friend circles
Segment Tree	Population pyramid — sum over any age range quickly
Fenwick Tree	Odometer with clever bit tricks for partial sums
LRU Cache	Small whiteboard — erase what was written longest ago for space
Bloom Filter	Bouncer list — might let someone in wrongly, never turns away regulars

7 Common Interview Questions

7.1 Arrays

- › Two Sum (LeetCode 1)
- › Best Time to Buy and Sell Stock (LC 121)
- › Maximum Subarray / Kadane's (LC 53)
- › Container With Most Water (LC 11)
- › Trapping Rain Water (LC 42)
- › Rotate Array (LC 189)
- › Product of Array Except Self (LC 238)
- › Merge Intervals (LC 56)

7.2 Strings

- › Longest Substring Without Repeating Characters (LC 3)
- › Minimum Window Substring (LC 76)
- › Group Anagrams (LC 49)
- › Valid Palindrome (LC 125)
- › Encode and Decode Strings (LC 271)
- › Longest Palindromic Substring (LC 5)
- › String to Integer (atoi) (LC 8)

7.3 Hash Maps/Sets

- › Two Sum (LC 1)
- › Top K Frequent Elements (LC 347)
- › Longest Consecutive Sequence (LC 128)
- › Subarray Sum Equals K (LC 560)
- › Valid Sudoku (LC 36)
- › Find All Anagrams in a String (LC 438)

7.4 Linked Lists

- › Reverse Linked List (LC 206)
- › Merge Two Sorted Lists (LC 21)
- › Linked List Cycle (LC 141)
- › Reorder List (LC 143)
- › Remove Nth Node From End (LC 19)
- › Merge K Sorted Lists (LC 23)

7.5 Stacks/Queues

- › Valid Parentheses (LC 20)
- › Min Stack (LC 155)
- › Daily Temperatures (LC 739)
- › Largest Rectangle in Histogram (LC 84)
- › Sliding Window Maximum (LC 239)
- › Evaluate Reverse Polish Notation (LC 150)

7.6 Trees

- › Maximum Depth of Binary Tree (LC 104)
- › Invert Binary Tree (LC 226)
- › Same Tree (LC 100)
- › Binary Tree Level Order Traversal (LC 102)

- › Validate Binary Search Tree (LC 98)
- › Lowest Common Ancestor of BST (LC 235)
- › Kth Smallest Element in BST (LC 230)
- › Serialize and Deserialize Binary Tree (LC 297)
- › Diameter of Binary Tree (LC 543)

7.7 Heaps

- › Kth Largest Element in an Array (LC 215)
- › K Closest Points to Origin (LC 973)
- › Find Median from Data Stream (LC 295)
- › Task Scheduler (LC 621)
- › Merge K Sorted Lists (LC 23)

7.8 Tries

- › Implement Trie (LC 208)
- › Word Search II (LC 212)
- › Replace Words (LC 648)
- › Design Search Autocomplete System (LC 642)

7.9 Graphs

- › Number of Islands (LC 200)
- › Clone Graph (LC 133)
- › Pacific Atlantic Water Flow (LC 417)
- › Course Schedule (LC 207) — cycle detection
- › Number of Connected Components (LC 323)
- › Word Ladder (LC 127) — BFS shortest path
- › Network Delay Time (LC 743) — Dijkstra
- › Cheapest Flights Within K Stops (LC 787) — modified Dijkstra/Bellman-Ford

7.10 Union-Find

- › Number of Provinces (LC 547)
- › Redundant Connection (LC 684)
- › Accounts Merge (LC 721)
- › Satisfiability of Equality Equations (LC 990)

7.11 LRU/LFU

- › LRU Cache (LC 146)
- › LFU Cache (LC 460)

7.12 Segment Tree / BIT

- › Range Sum Query — Mutable (LC 307)
- › Count of Smaller Numbers After Self (LC 315)
- › Reverse Pairs (LC 493)

8 Typical Mistakes Candidates Make

1. **Using `list.pop(0)` for queues** — It's $O(n)$. Always use `collections.deque` and `popleft()`.
2. **String concatenation in loops** — `result += s` inside a loop is $O(n^2)$. Use `"".join(parts)`.

3. **Not handling the empty/single-element edge case** — For linked lists, trees, arrays. Check if not root, if not nums, if `len(nums) < 2` early.
4. **BST validation using only local `left < root < right`** — This misses nodes in subtrees. Pass bounds (`lo`, `hi`) down the recursion.
5. **Modifying a collection while iterating it** — Causes `RuntimeError`. Copy or collect indices first.
6. **Forgetting to mark nodes visited before enqueueing (BFS)** — Leads to exponential revisits. Mark visited when you enqueue, not when you dequeue.
7. **Using a set as a hash map key** — Sets are unhashable. Use `frozenset` or `tuple(sorted(...))`.
8. **Forgetting Python's `heapq` is min-heap only** — For max-heap, push `-val` and negate on pop.
9. **$O(n^2)$ nested loop when $O(n)$ hash map exists** — When you write two nested loops looking for a pair, always ask: "can I precompute what I'm looking for?"
10. **Not using dummy head node for linked lists** — Head deletion becomes a special case. Dummy head eliminates it.
11. **Recursion without memoization** — Fibonacci, path counting, etc. naturally revisit sub-problems. Add `@lru_cache` or a memo dict.
12. **Union-Find without path compression** — Without it, find is $O(n)$. Always compress.
13. **Assuming dict maintains insertion order in Python 2** — It doesn't. Python 3.7+ guarantees it; don't rely on this in cross-version code.
14. **Off-by-one in sliding window** — When window is `[l, r]`, size is `r - l + 1`. Sketch a small example before coding.
15. **Passing arrays by reference and mutating them** — In Python lists are passed by reference. If you need to preserve the original, copy with `arr[:]`.

9 Revision Cheat Sheet

9.1 Complexities to Memorize (Absolute Must)

Array access:	$O(1)$	Hash map get/put:	$O(1)$ avg
Array search:	$O(n)$	BST search:	$O(\log n)$ balanced
Append to list:	$O(1)$ A	Heap push/pop:	$O(\log n)$
List insert/del:	$O(n)$	Build heap:	$O(n)$ ← non-obvious!
Trie ops:	$O(m)$	Segment tree:	$O(\log n)$ query/update
BFS/DFS:	$O(V+E)$	Union-Find:	$O(\alpha(n)) \approx O(1)$
Dijkstra:	$O((V+E)\log V)$		
Heapsort:	$O(n \log n)$		
Sort:	$O(n \log n)$		
Binary search:	$O(\log n)$		
String join:	$O(n)$	String concat loop:	$O(n^2)$ ← DANGER

A = amortized

9.2 Decision Hooks (6 questions to ask every interview)

1. **"Do I need $O(1)$ lookup?"** → Hash map or hash set.
2. **"Do I need sorted order or range queries?"** → BST, sorted array + binary search, or segment tree.
3. **"Do I need repeated min/max?"** → Heap.
4. **"Is this a traversal/reachability problem?"** → BFS (shortest path) or DFS (connectivity, cycles).
5. **"Am I connecting/grouping things?"** → Union-Find.
6. **"Do I need prefix operations?"** → Trie (strings) or Fenwick tree (numbers).

9.3 Quick Pattern Mapping

Pattern	Structure
Sliding window	Array + two pointers
Sliding window max/min	Monotonic deque
Next greater element	Monotonic stack
Level-order traversal	Queue (BFS)
Path/exhaustive search	Stack (DFS) / recursion
Shortest path (unweighted)	BFS
Shortest path (weighted)	Dijkstra (min-heap)
Connected components	BFS/DFS or Union-Find
Top K elements	Heap of size K
Frequency counting	Hash map / Counter
Deduplication	Hash set
Prefix sum	Fenwick tree / prefix array
Range sum (mutable)	Fenwick tree
Range min/max (mutable)	Segment tree
Word prefix search	Trie
Order + O(1) eviction	LRU (HashMap + DLL)
Large-scale membership	Bloom filter

9.4 The 5 Structures You MUST Be Able to Code From Scratch

1. **Hash Map** (open addressing or chaining — at least explain internals)
2. **Linked List** with insert, delete, reverse
3. **Binary Heap** with push, pop, heapify
4. **Trie** with insert, search, startsWith
5. **LRU Cache** (HashMap + doubly linked list)

9.5 Memory Aids

- › **Heap is NOT a BST** — heap has parent $\leq$ children (only), BST has left $<$ parent $<$ right.
- › **Trie depth = key length, not n** — n = number of keys.
- › **Union-Find has no delete** — it's append-only by design.
- › **BFS finds shortest path; DFS does not** — in unweighted graphs.
- › **heapify is O(n)** — this surprises everyone. Proof: most nodes are at the bottom and need to sift down only a little.
- › **Hash map worst case O(n)** — when all keys hash to same bucket (adversarial input or bad hash). In practice O(1).
- › **AVL is stricter than Red-Black** — AVL: max height diff 1; RB: height $\leq 2 \cdot \log(n)$. RB faster for insert/delete, AVL faster for lookup.

10 Visual Reference

A set of figures distilling the key quantitative intuitions of this topic.

Data structure operation complexity (average case)				
	Access	Search	Insert	Delete
Array	$O(1)$	$O(n)$	$O(n)$	$O(n)$
Dyn. Array	$O(1)$	$O(n)$	$O(n)$	$O(n)$
Linked List	$O(n)$	$O(n)$	$O(1)$	$O(1)$
Hash Table	$O(1)$	$O(1)$	$O(1)$	$O(1)$
BST (bal.)	$O(\log n)$	$O(\log n)$	$O(\log n)$	$O(\log n)$
Heap	$O(n)$	$O(n)$	$O(\log n)$	$O(\log n)$
Stack/Queue	$O(1)$	$O(n)$	$O(1)$	$O(1)$

Figure 1. Pick the structure that makes your hot operation $O(1)/O(\log n)$. Hash tables win for lookups; balanced trees keep order; heaps give cheap min/max.